

(a) Decompose $\rightleftharpoons$ compose $\rightarrow$ data flow $\dashrightarrow$ score / credit $\boxed{u}$ rejected

Masked loop

 $\mathcal{L} =$
 (Init, B , S) $\rightarrow$
 $\mathcal{G}$ hidden

within a rollout

1 Rollout
 y_1 $\begin{bmatrix} a & b & c & a & b \end{bmatrix} \checkmark$
 y_2 $\begin{bmatrix} c & u & d & e & c \end{bmatrix} \times$
 y_3 $\begin{bmatrix} v & d & u & e & f \end{bmatrix} \times$
2 Decompose
 A_1 $\begin{bmatrix} a & b & c & a & b \end{bmatrix}$
 A_2 $\begin{bmatrix} c & u & d & e & c \end{bmatrix}$
 A_3 $\begin{bmatrix} v & d & u & e & f \end{bmatrix}$

across rollouts

3 Pool
 $\begin{bmatrix} a & b & u \\ c & d & v \\ e & f & \end{bmatrix}$
4 Filter
 $\begin{bmatrix} a & b & c \\ d & e & f \end{bmatrix}$
5 ComposeOutput
 C_{surv} $\downarrow$ composed set $\downarrow$ rollout group**(b) SFT**compose $\rightarrow$ label

Composed set

 $C_{\mathcal{L}}^{(t)}$

resample

Coverage $\geq \tau$?

yes

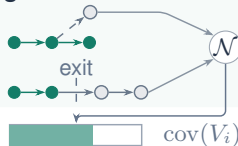
SFT labels

Fine-tune

(c) Target-free scorer**Binary** valid set $\mapsto \{0, 1\}$ **Candidate negatives**

off-trajectory

post-exit

**Coverage****Shapley****allocation**
 y_1 $\begin{bmatrix} \otimes & \otimes & \otimes & \circ & \circ \end{bmatrix} \phi_1$
 y_2 $\begin{bmatrix} \otimes & \otimes & \otimes & \circ & \otimes \end{bmatrix} \phi_2$
 y_3 $\begin{bmatrix} \circ & \circ & \otimes & \bullet & \otimes \end{bmatrix} \phi_3$
 $\bullet$ unique $\otimes$ shared
Full reward per rollout

$$r_i = \text{cov}(V_i) + 0.3\phi_i$$

(d) RL

attribute exploration

 y_1 $\begin{bmatrix} a & b & c & a & b \end{bmatrix}$
 y_2 $\begin{bmatrix} c & u & d & e & c \end{bmatrix}$
 y_3 $\begin{bmatrix} v & d & u & e & f \end{bmatrix}$

pool, filtered once

 $\begin{bmatrix} a & b & c & d & e & f \end{bmatrix}$
 r_1 r_2 r_3
GRPO
 group-normalized
 advantage
 $\rightarrow$ policy update